VideoWorld2-IDM CALVIN Gate Status Report

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2026-04-26

Verdict

The current repository supports an offline CALVIN gate and a mock-only closed-loop smoke path. It still does not contain a valid real CALVIN closed-loop adjudication. The real closed-loop path is blocked by two concrete facts: the repository evaluator rejects non-mock datasets, and the local rescue bundle does not contain raw CALVIN frame/action files needed by the dataset loader.

Repository Overview Figure

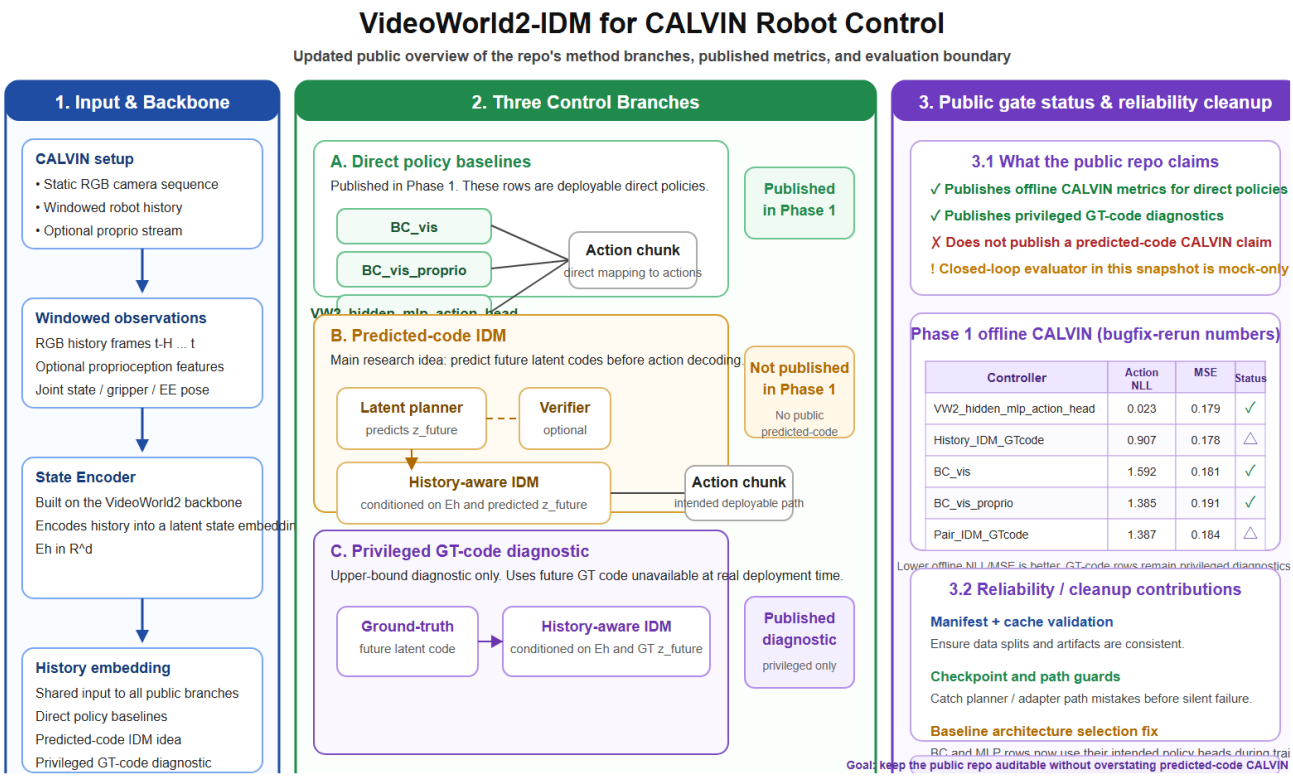

Figure 1: Public repository overview refreshed to match the current committed bugfix-rerun metrics and status boundary. The original user-provided draft was not published verbatim because its Phase 1 table contained stale numbers.

Reliability Changes

- The latent planner now uses a causal decoder mask, preventing teacher-forced target tokens from leaking future labels into earlier predictions.
- Latent caches and window indexes now carry validation metadata and content digests; saved window payloads are re-derived before reuse, and stale specs, stale adapter checkpoint contents, stale split identity, stale full-span CALVIN frame provenance, duplicate cache keys, or key-set mismatch now fail closed.
- Official dLDM tokenizer checkpoints are rejected unless they cover the encode path used for cache extraction and predicted-code embeddings.
- Train and validation manifests are checked for shared episode IDs, shared file-backed episode sources, byte-level and semantic copied file contents, copied CALVIN frame contents across train/validation, duplicate copied CALVIN frame contents under different roots within a split, and train/validation CALVIN frame-span overlap before dataset construction. Same-root within-split CALVIN span overlaps are recorded as annotation-window weighting.
- DataLoader workers and generators are seeded from the experiment seed, and evaluation entrypoints request deterministic CUDA behavior by default.
- Direct-policy, MLP, planner, verifier, teacher, resume, and CLI checkpoints are validated against explicit architecture, IDM conditioning metadata, planner/teacher checkpoint identity, and path guards before loading; distillation now preserves predicted-code teacher conditioning.
- Verifier-assisted evaluation reloads the verifier's paired state encoder instead of feeding a verifier trained on one encoder with tokens from another encoder.
- Evaluation and diagnostic JSON writers reject non-finite metrics before publication, and verifier reranking applies the jerk penalty per candidate instead of using a batch-global scalar.
- Cross-directory config inheritance preserves the directory that defined each relative path, and repeated relative path strings are resolved by config key path to avoid source-directory collisions.

Phase 1 Offline CALVIN Metrics

The table below reflects the current committed `results/phase1_offline_metrics.json` values. It supersedes the earlier stale public table after rerunning Phase 1 with the adapter-path fix and the baseline architecture-selection fix. This remains an offline gate over the configured validation latent-cache subset, not a real CALVIN closed-loop claim. Action NLL and action MSE are raw-vector losses over the stored CALVIN action representation, and offline jerk is computed on the predicted action chunk rather than on a real executed closed-loop stream.

Controller	Action NLL	Action MSE	Jerk	Interpretation
VW2_hidden_mlp_action_head	0.02332	0.17899	0.00356026	direct MLP policy
History_IDM_GTcode	0.90677	0.17816	0.00000010	privileged GT-code upper bound
BC_vis_proprio	1.38503	0.19142	0.00000064	direct visual+proprio policy
Pair_IDM_GTcode	1.38689	0.18394	0.00000206	privileged GT-code upper bound
BC_vis	1.59221	0.18078	0.00000014	direct visual policy

The GT-code rows consume future latent codes from the target trajectory. They are useful diagnostic upper bounds, but they are not deployable policies. The deployable rows in this table are the direct-policy baselines and the MLP action head. The public JSON/CSV set `planner_code_accuracy` to zero for these rows because no predicted-code planner result is published in this snapshot.

Phase 0 Mock Smoke

Check	Result	Interpretation
Oracle replay	100% success on 24 validation episodes	synthetic mock oracle diagnostic
BC overfit closed loop	33.33% success, offline MSE 0.00283	synthetic mock smoke policy
History-IDM GT-code overfit closed loop	16.67% success, offline MSE 0.00346	privileged future-code mock diagnostic

These smoke runs validate plumbing on a synthetic mock environment. They do not prove real CALVIN closed-loop behavior. The repository also includes machine-readable mock-only rollout provenance in `results/phase0_rollout_metadata.json`; the public summary JSON duplicates the GT-code privilege flag inside the affected rows and leaves planner-code accuracy null when no predicted-code planner is evaluated.

Artifact Boundary

The non-versioned rescue bundle contains manifests, window indexes, latent caches, checkpoint files, resolved configs, and offline-evaluation outputs. It does not contain raw CALVIN RGB/proprio/action frame files. The rescued manifests still reference the original remote Linux roots, so local replay requires remapping or restoring the raw dataset. The conditional real closed-loop goal remains open until the real CALVIN environment and raw data are available within the run budget policy.

The locally observed `ldm_tokenizer_training_init_weights.pt` file is not accepted as a verified official tokenizer for fresh cache extraction because it does not cover all encode-path parameters. Fresh official-cache extraction requires a complete tokenizer checkpoint, or an explicitly diagnostic partial-checkpoint override.

The committed `results/rescued_artifact_audit.json` records that rescued train and validation window payloads match the current window builder: 2,102 train windows and 12,881 validation windows. It checks latent-cache `(episode_id, t)` keys against the current train-window builder and the configured validation-window prefix. It also records same-root within-split CALVIN span overlaps, which are interpreted as annotation-window weighting rather than independent raw-frame evidence. Because the rescued manifests reference remote Linux roots and raw frames are absent locally, this audit does not prove byte-level raw-frame content separation for those remote roots.

Public Files

The public result tables and rescued-artifact audit are committed under `results/`. The architecture diagrams are committed as source Mermaid files under `docs/architecture/` and rendered images under `assets/readme_figs/`. Stale report copies and unused demo assets were removed from the public repository.